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CT60A5401 GAME DEVELOPMENT PROJECT

ACADEMIC YEAR 2022-2023

ASSOC. PROF. JUSSI KASURINEN (D.SC.)

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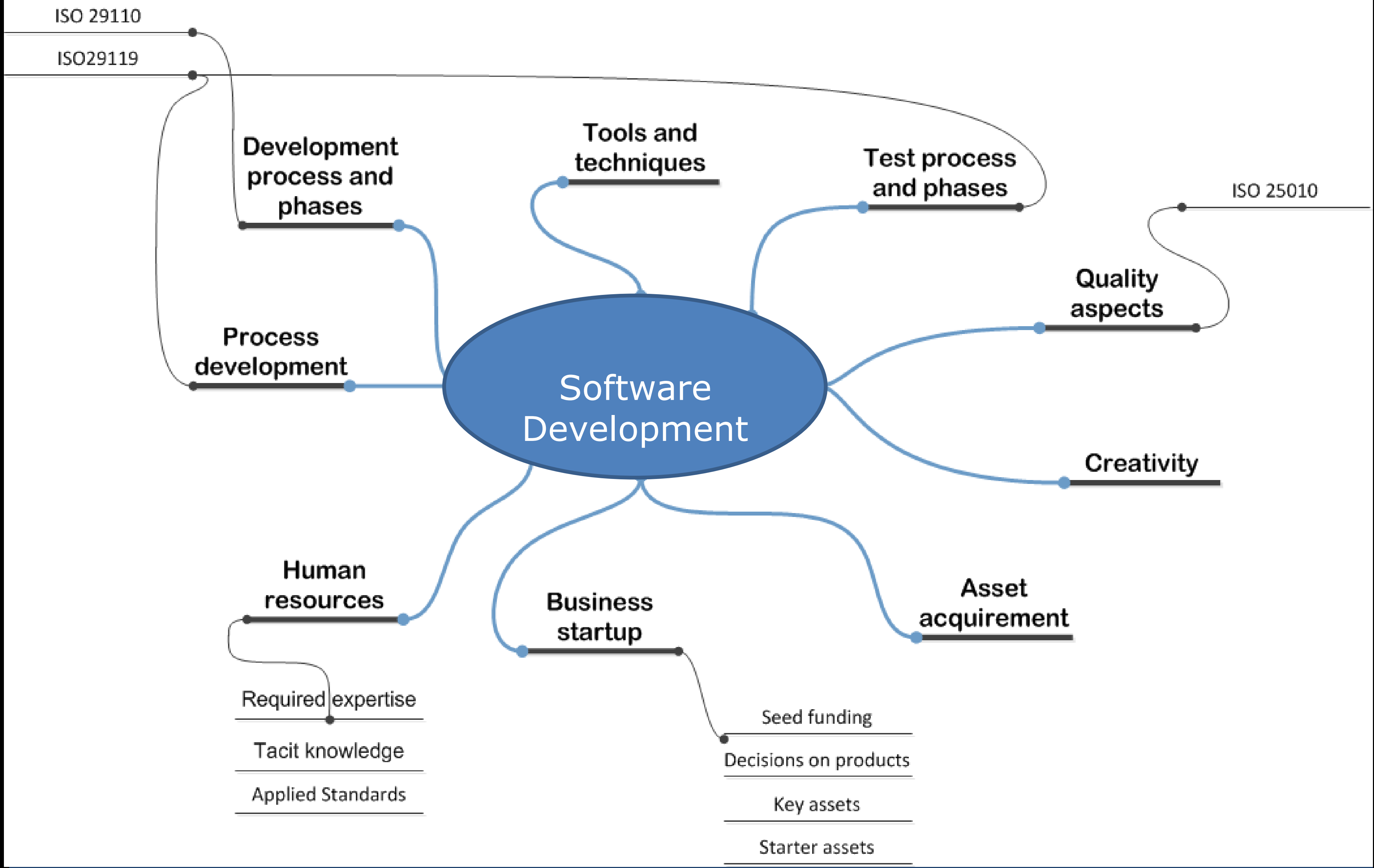
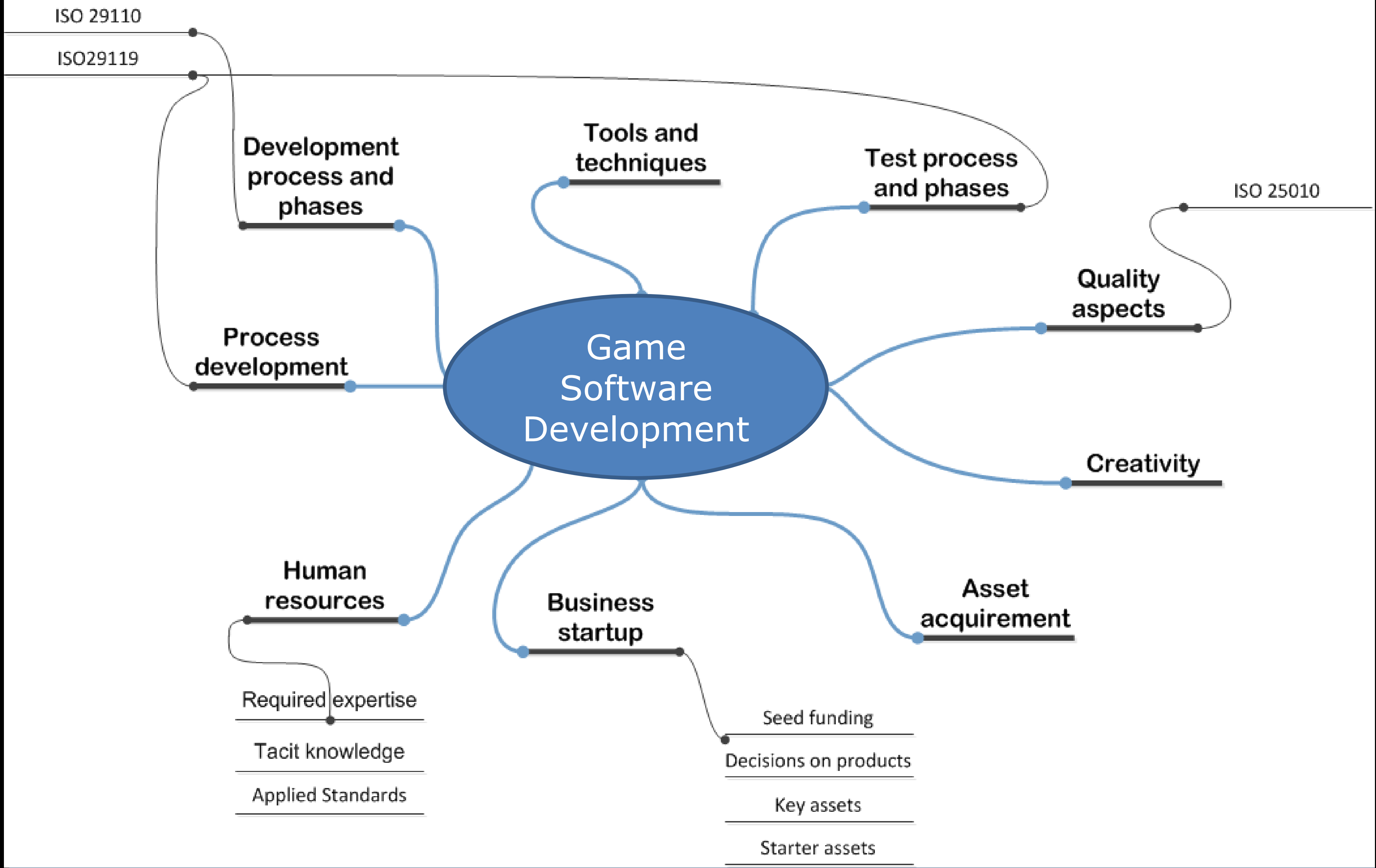
TODAY’S AGENDA

- Games from the viewpoint of software engineering


ISO 29110
ISO29119
Development process and phases
Tools and techniques
Test process and phases
ISO 25010
Quality aspects
Process development
Software Development
Creativity
Human resources
Business startup
Asset acquirement
Required expertise
Tacit knowledge
Applied Standards
Seed funding
Decisions on products
Key assets
Starter assets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Making games

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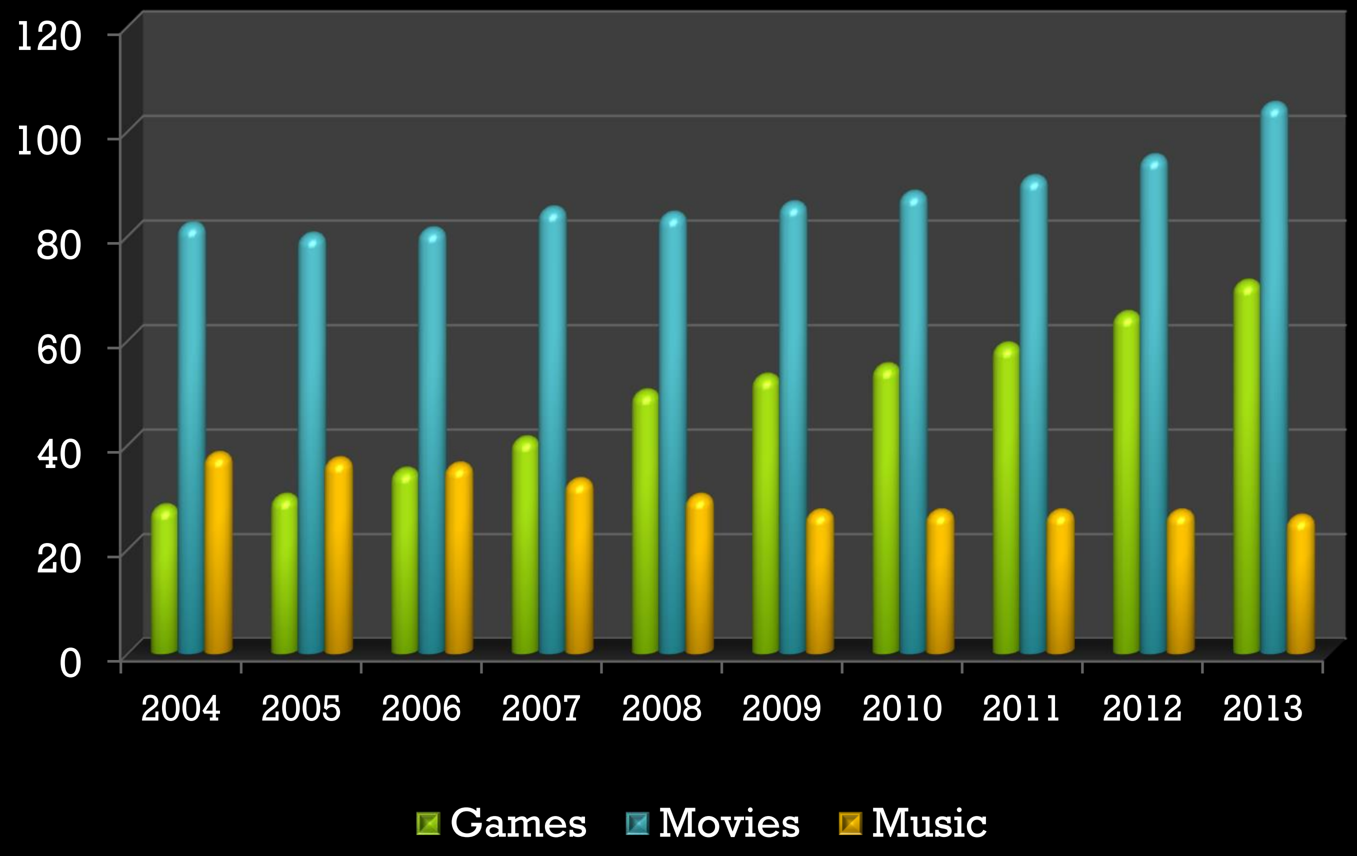
Making money



Global sales in billions of USD

PriceWaterhouseCoopers, Global Media Outlook 2008-2013

Entire Software Industry, 303.8 B (in 2007)

Player demographics

- 72 % of households play games
- Average age of player: 37 years
- 42 % gamers are female.
- 33 % of gamers, games their favorite hobby.

Entertainment Software Association (ESA), 2011 report on game industry demoghraphics in USA

Software development

- Interestingly, game development and software development share some common features.
- Design work
- Development, programming tasks
- Tools
- Testing
- Objectives:
  - In time
  - In budget
  - With the intended features
  - With acceptable quality

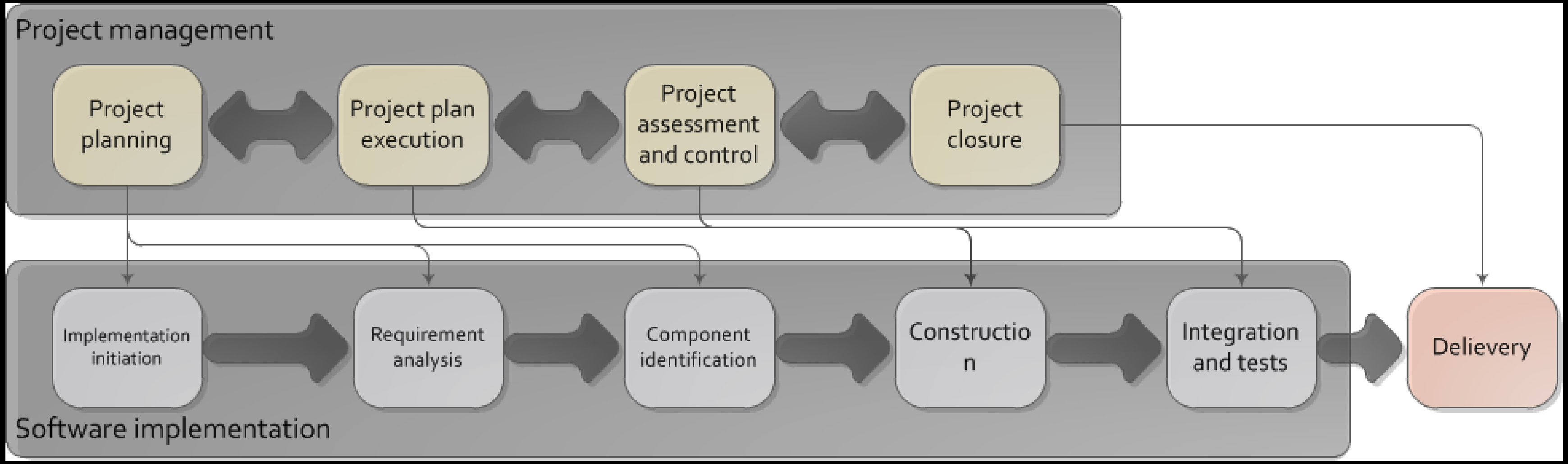
Software development

- Obviously there also are some differences.
- More emphasis on customers, market
- Design with artistic vision on how things happen.
- Tools not necessarily the same as in software dev.
- Testing with users.
- Sound design.
- Development has certain aspects which are unusual:
  - Real time client-to-client multiplayer systems
  - Artificial intelligence
  - 3D mathematics
- Objectives:
  - Is the game fun?

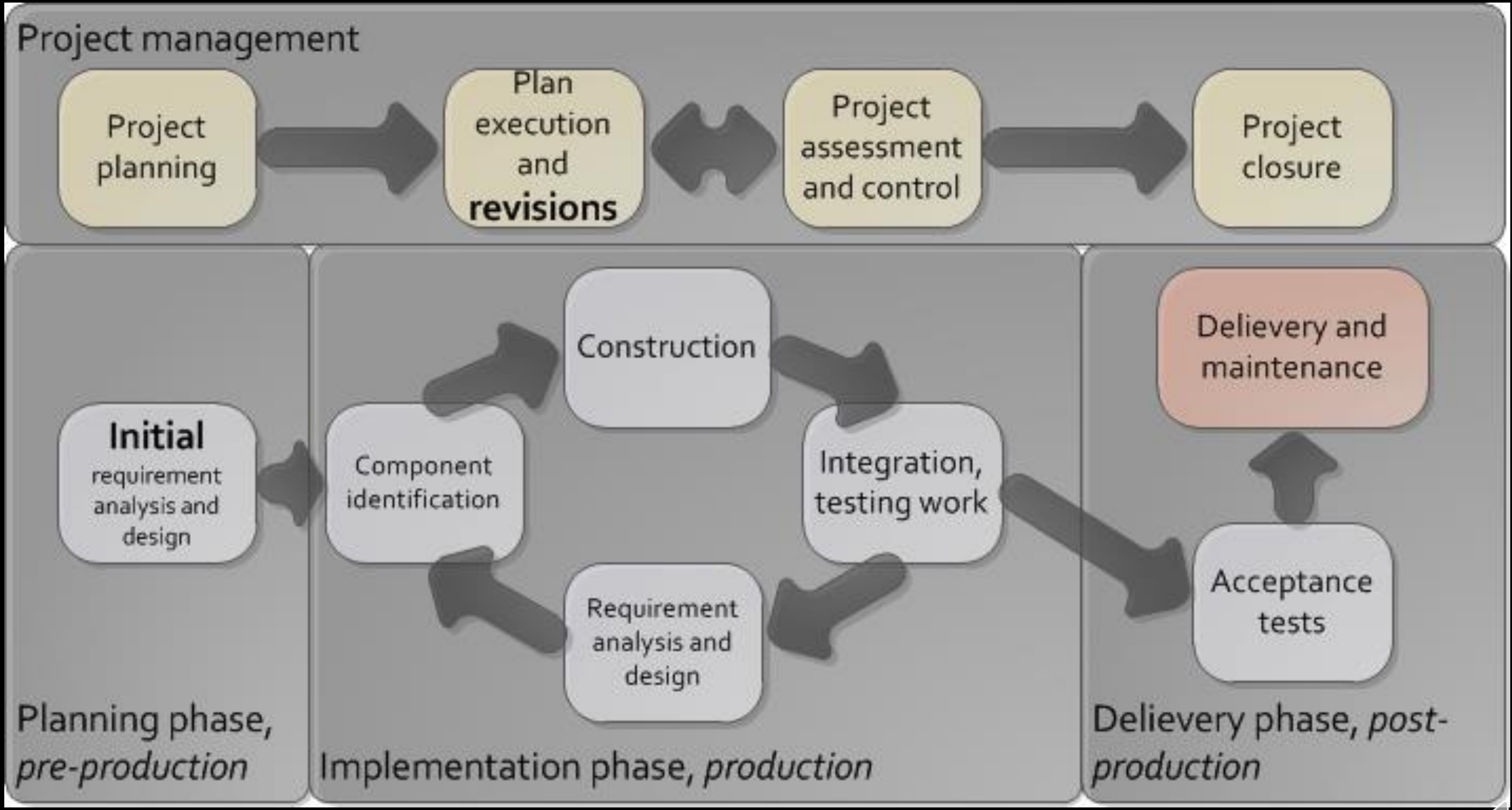
Existing Knowledge

- One problem also is that we have extensive knowledge on how software should be built.
  - Can this work with games?

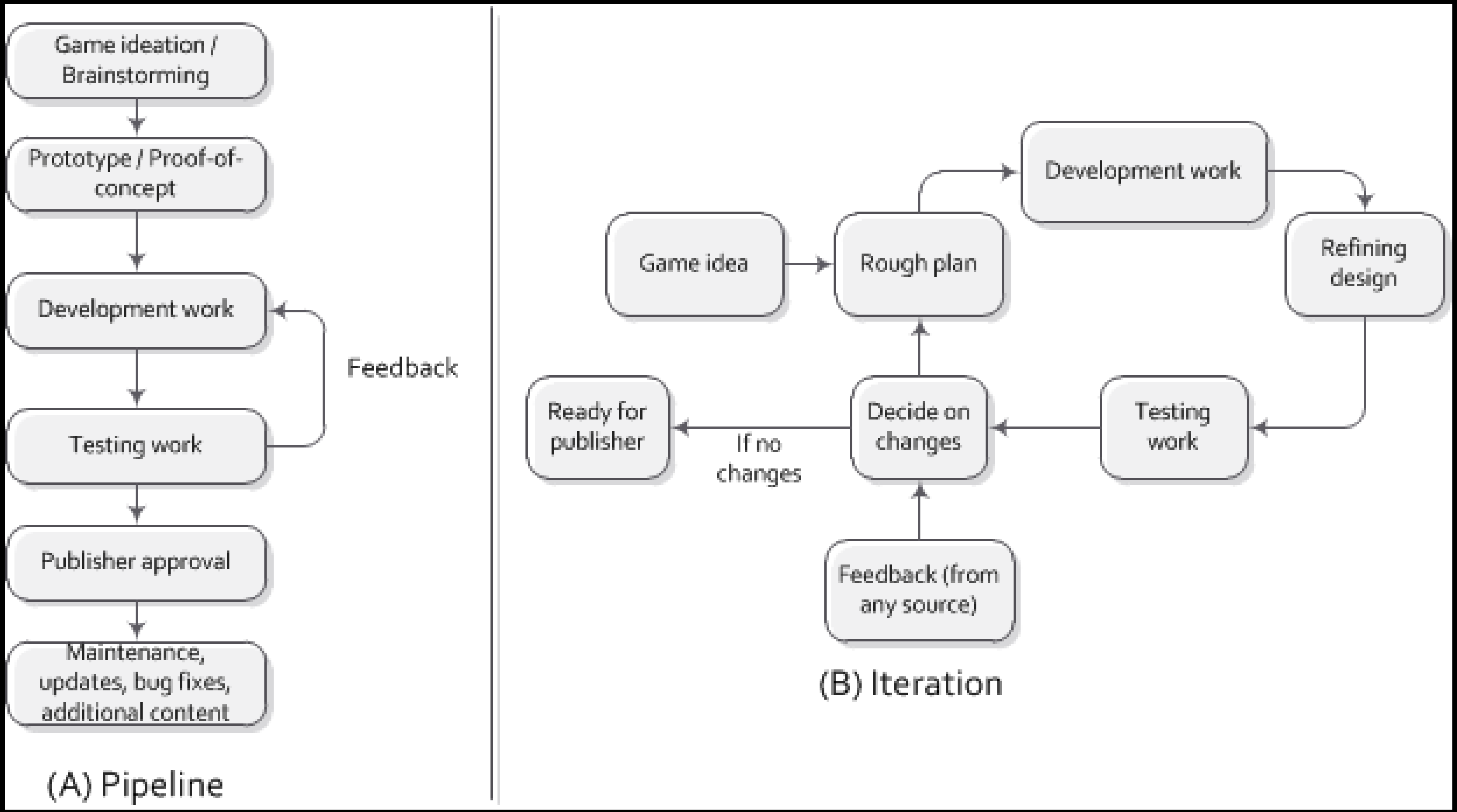
For example, ISO/IEC 29110 here:



Unfortunately, no.



Developing games in practice



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Testing

- Unlike software industry, game testing usually focuses on user experience.
- Testing work for technical aspects is always present, but not necessarily the main point.
- Also, content testing, as in internal logic, features, for example ensuring that there are no ”always wins”-strategies.
- Amount of testing-dedicated resources similar, if not even better, than with traditional software industry.

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On Testing

- Most usual ways of doing testing work:
  1. User-testing
  2. Usability testing
  3. Explorative testing
  4. Unit testing (and everything related to unit testing)
- Objectives of testing:
  - Technically: “Good enough is enough.”
  - Mechanically: “Make sure there is no way to always win.”
  - User Experience: “Is this what we want?”

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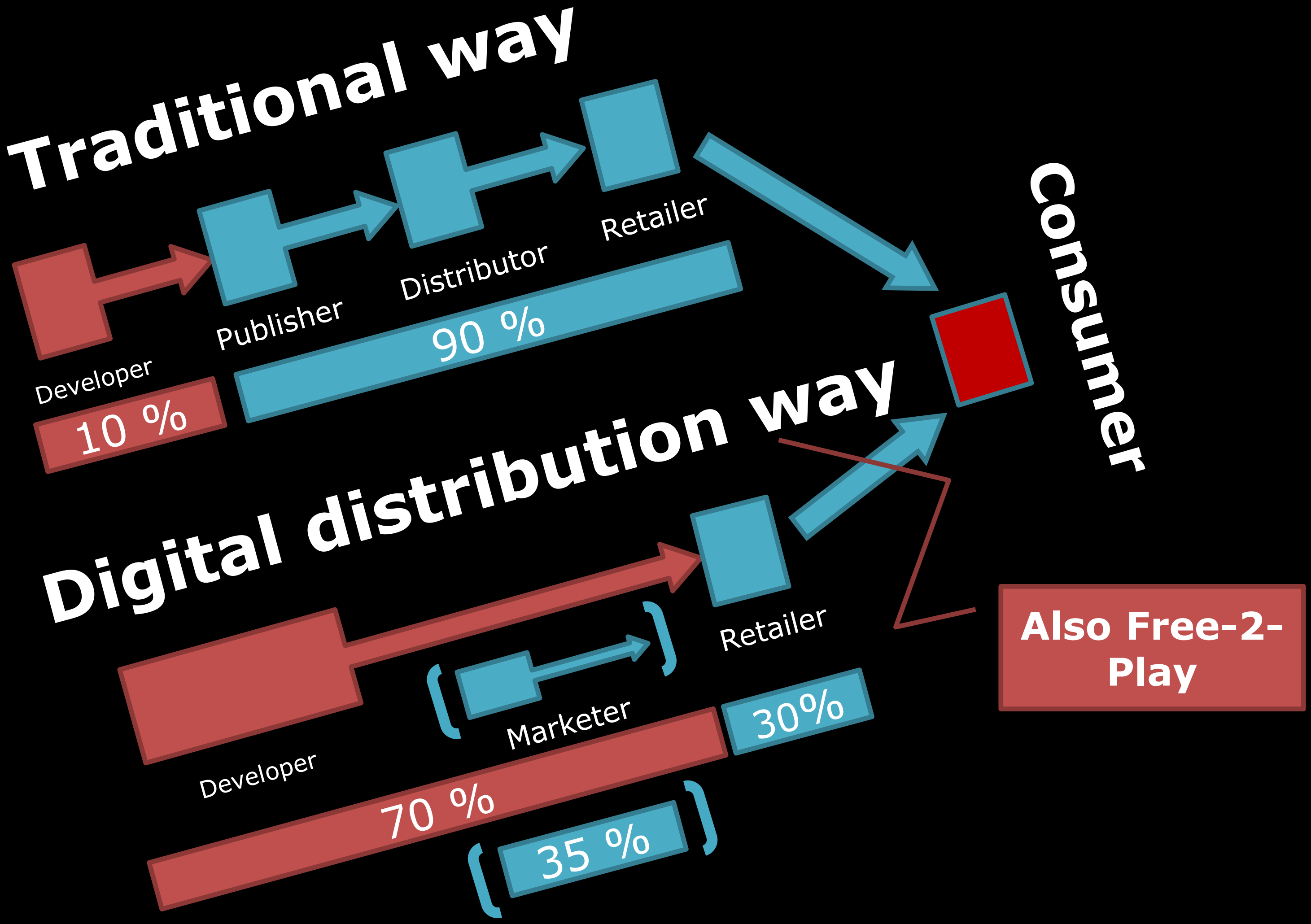
Tools

- As for expectations and needs, it seems that *game developing companies expect their tools to allow easy prototyping and ability to design while implementing; large parts of the design is done with same tools as the actual development work.*
- Most common of the few mentioned issues were related to amount of bugs, or user interface design.
- Some organizations also considered that if they would have to increase the number of people in the project, they would probably have to adopt some sort of formalized project management system or more systematic development process model.
- The case organizations did not apply self-made tools to a large degree. In fact, the case organizations were immigrating away from self-build tools towards third party systems.

Business Logistics



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(Pay-2-Play) Business in numbers

- Profit for non-marketed game, released at App store: 74 euros.
- Profit for marketed game: 25000+ euros.
- Worth of being in TOP 25: 5000 euros per day.
- Mobile games take 3-6 months to make.
- 90% of sales within 1 month from release.
- By default international business.
- Free-2-Play models change everything.

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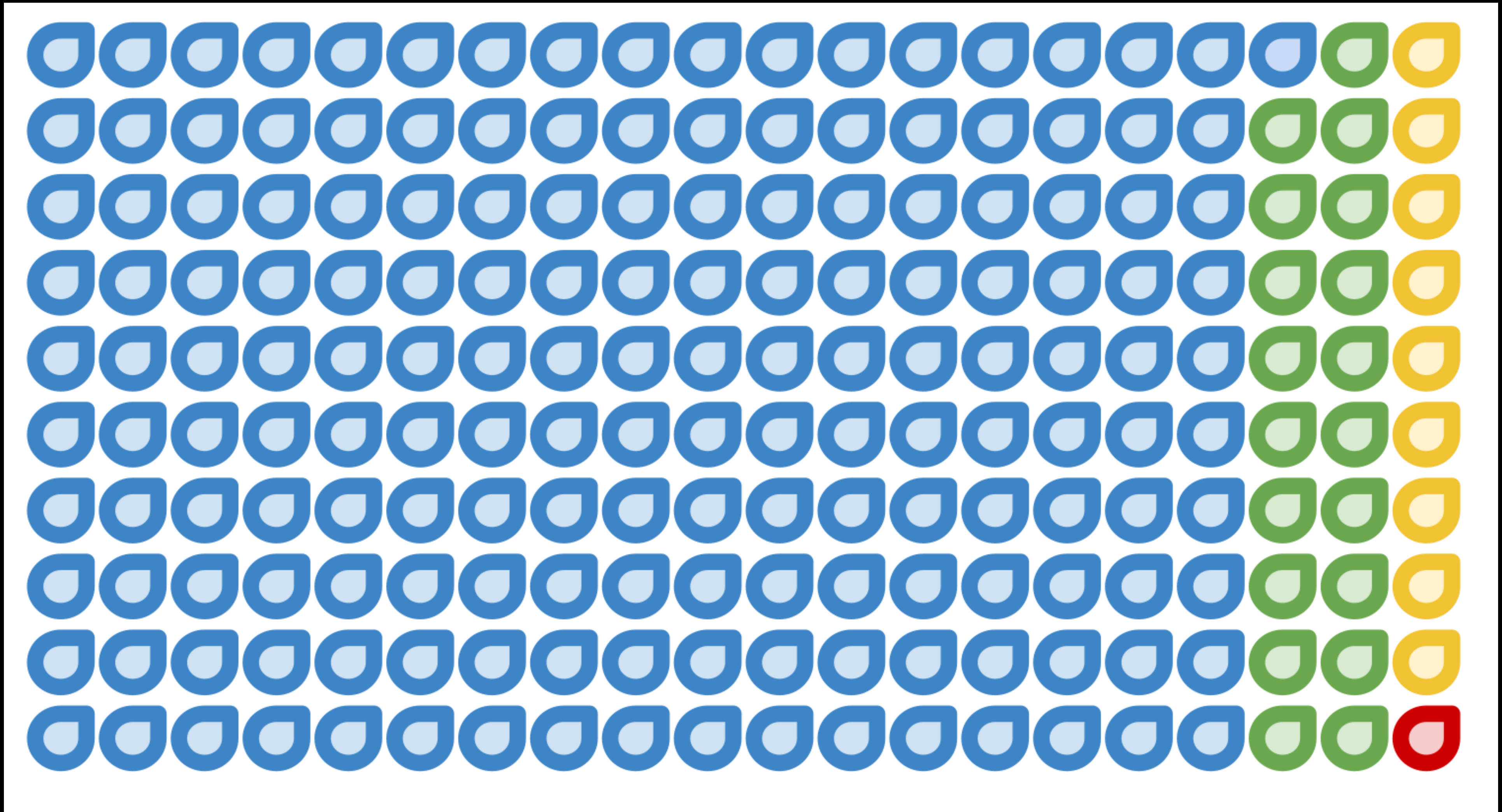
Business in numbers (Free-2-Play)

- Marketing is basically purchasing users (For each X Eur. spent we get Y new users); The people who download and start your game at least once. Out of those….
- (Up to) 14 % of users might play your game more than once.
- 95 % of players never pay for anything
- 4 % of players might buy something once or twice. (Or pay for something extra.)
- 1 % of players are the ”whales”; They provide constant stream of income, even thousands of dollars per user.
- **Money from maintenance, continuous development of new content, “live-ops”**

Business in numbers (Free-2-Play)



+ those users who saw the advertisement, but did not click it, or did not install your game, or installed and never bothered to try it out.

Game business in 5 concepts (lifted from presentation by Rovio)

1. Mobile business
2. Free-2-Play
3. In-app purchases
4. Whales
5. Live-Ops

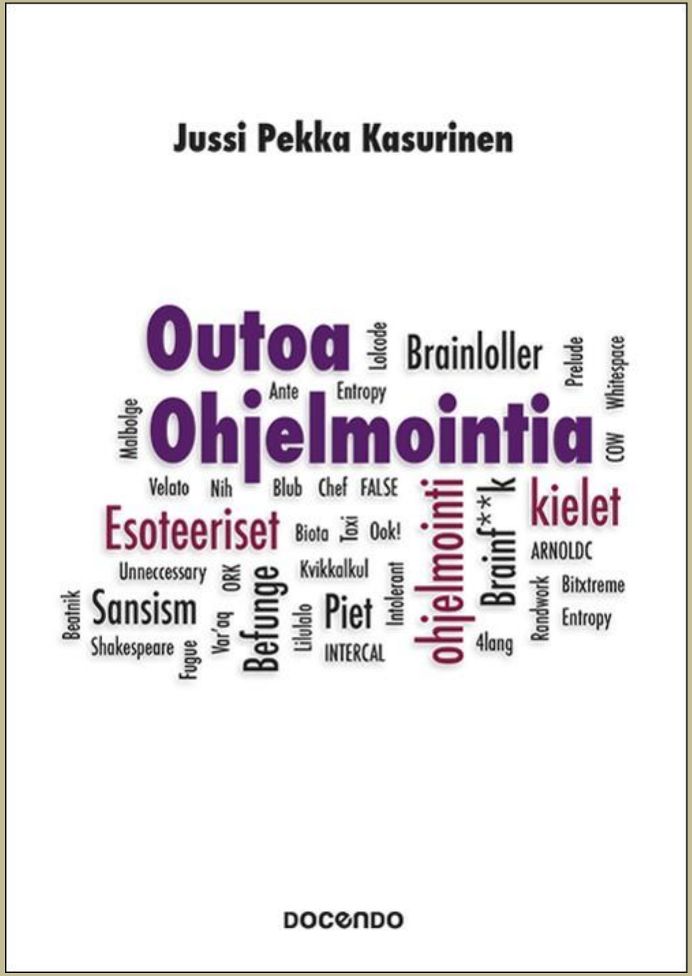
Visibility

Short list of things to know about your 1st product

- Nobody cares about what you are doing.
- No-one will try out your new idea/game/app.
- No-one will pay for the app you are making.
- Nobody wants to invest in your team.
- No-one knows you exist.

This is the starting point.

Visibility. An example.



Book about Computer Science Fundamentals, using esoteric languages (including brainf*ck, as examples)

brainf*ck

Back to the point

On innovation

- Dominant systems to the degree of standardization.
- Sequel after sequels after sequels after…
- ”From ’creative chaos’ to process thinking”… ”creativity as a publicity stunt.”
- Multiple genres: One product cannot fulfill all customer needs.
- Themes, trends change, what sells today is old news tomorrow.
- Innovation, creativity needed in the development of new products, merchandises.

0,1 % marketshare == 73 000 000 USD (in 2013)

On Game Design

- Games deliver user experience.
- Playability, having fun while playing important.
- Technical viewpoint, “good enough” is enough.
- Game development: “Creative work with software engineering stuff in it.”

On Game Design

- Games are not art for art’s sake ::
  Games are usually designed with profit in mind
- Artistic merits and creativity secondary, although well-received attribute.
- Nevertheless, most of the people in game business consider games to be more creative than software.
  - Profit vs. Merit → Almost always profit.
- Usual ways to design games::
  Brainstorming, vision, pitching contests, prototyping, pen and paper.
- Almost always marketing and testing involved :: ***”What is fun?” or ”Is this game fun?”***

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Normal startup team

- Mobile games for smartphones or tablets.
- 2-4 people, work as owners without salary: usually designer, programmer and artist.
- Everybody does everything, but everyone has one or few particular specialities.
- Simple game design to get something published → "Big players" only interested in teams that have proof that they can deliver projects.
- Biggest expenses tools, marketing.
- Sound effects, secondary 3D models, music outsourced.
- Three most important problems named by the startups:
  1. Money
  2. Money and
  3. Money.
- If you have money, everything else is just elbow grease and sleepless nights.

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Things That Matter

Rank	Element	Weights						
		Case A	Case B	Case C	Case D	Case E	Case F	Total
1	Human capital	0.365	0.318	0.267	0.265	0.350	0.317	0.314
2	Marketing	0.205	0.085	0.035	0.114	0.202	0.209	0.142
3	Financing	0.075	0.203	0.135	0.135	0.056	0.107	0.118
4	Key partners	0.070	0.157	0.185	0.089	0.112	0.068	0.113
5	Customer relationship	0.058	0.050	0.099	0.235	0.122	0.091	0.109
6	Key resources	0.137	0.039	0.086	0.027	0.024	0.038	0.059
7	Key activities	0.031	0.075	0.095	0.042	0.035	0.062	0.057
8	Innovation process	0.038	0.056	0.055	0.054	0.030	0.086	0.053
9	Customer segment	0.020	0.017	0.042	0.040	0.069	0.022	0.035

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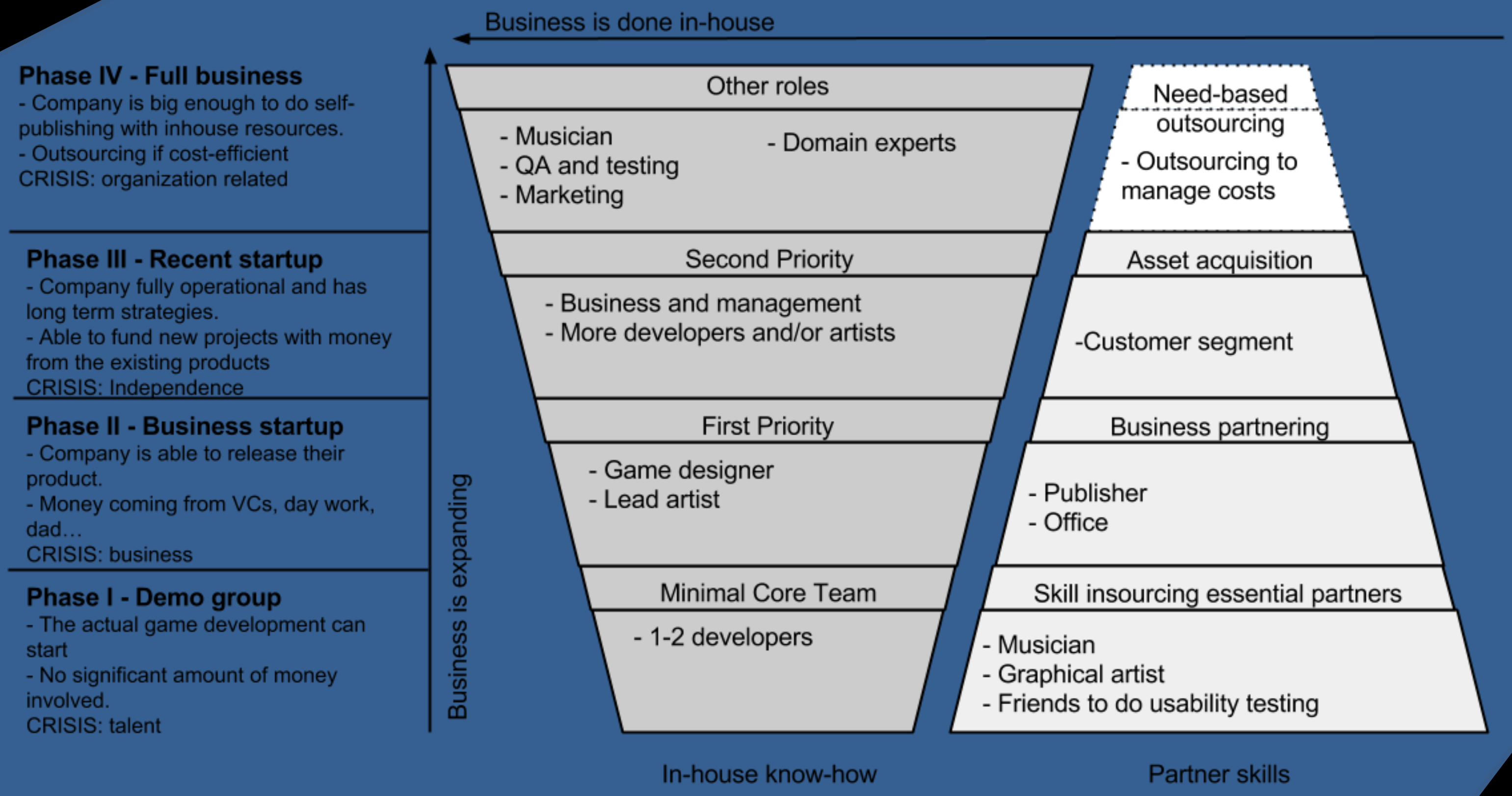
Things that matter: What companies want? (Collected from Supercell, Rovio)

- Actual talent is rare.
- Passion to games, not to ”having a career in games”
- Learn the fundamentals (C++, mathematics, physics)
- No use for mediocre talent; focus on your best skills.
- Do several projects, several themes, several genres.

Development Model for Startups



Actual Research papers:

- Technical framework and tools of game development.
- Applicability of the existing software models, especially things like ISO/IEC 29110.
- How game industry adopts new technology, especially cloud services and cloud systems.
- How games are designed.
- How game development should be taught to the computer science students.
- How gamification user interface can be tailored to enhance usability and motivational aspects.
- How user motivation can be detected from their activities.

Actual Research papers:

- How games are developed.
- How games are tested.
- How game startup companies evolve as they mature.
- What are the important elements of the company business-wise?
- How business models could be used to develop business?

REST OF THE COURSE

1. **Ensure that following things are online:**
   - Your game
   - Original GDD
   - Game source code
   - Your game trailer video
   - Tutorial projects + their video
2. **Industry viewpoint lecture week 45, last lecture in December (see schedules at Moodle).**

REST OF THE COURSE

- Materials have to stay online until teacher says it is OK to remove them (via Moodle).
- No materials = No grade
- No permission to view = No grade
- No review = No grade
- Deadline missed = No grade
- See Moodle for more details.

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Comments, feedback, want to call the lecturer a c*?**

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